

I Love You, Quite Literally

An Analysis of Terms of Endearment in a Dyadic Communication System

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ABSTRACT

This study examines the chat history between Mr. Sangchun (Harry) Yeh and Ms. Michelle Yang over the timespan from June 2025 to May 2026. It introduces statistical methods to analyze the subjects' usage of terms of endearment from lexical, temporal, and linguistic dimensions. The results show consistent fluctuations in communication frequency patterns and language selections. The study concludes that even when their surroundings undergo changes, both Subject A and Subject B continue to express affection in a sustained and adaptive manner.

Keywords

Affectionate Language; Code-switching; Dyadic Communication; Healthy Relationships; Longitudinal Analysis; Python; Textual Analysis; Terms of Endearment

1. INTRODUCTION

On May 31st, 2025, Mr. Sangchun (Harry) Yeh and Ms. Michelle Yang permanently altered their relationship titles from platonic peers to significant others. From that day, the regular dyadic communication between Yeh and Yang witnessed a fast development in the quantity and variety of terms of endearments¹ with which Yeh and Yang frequently refer to one another.

Yeh and Yang's usage of such terms of endearment offer an interesting insight into their interpersonal partnership, as such terms vary in denotation, language, and usage frequency. This study, thus, is interested in analyzing the patterns within Yeh's, referred to as Subject A, and Yang's, referred to as Subject B, communication system and their use of terms of endearment over the first year of their amatory collaboration.

2. RELATED WORKS

Prior research has introduced the method of analyzing chat histories between subjects to observe their relationship. Such frameworks have examined communication systems from the perspectives of frequency, commonly used phrases and terms, and high-level activity hours.

This study differs from past analysis frameworks by introducing a third dimension: language. The subjects of this study are both multilingual and utilize extensive intrasentential code-switching² in their habitual conversations. This study, thus, proposes a new framework in which the subjects' relationship is evaluated through their language choices and changes over time.

3. METHODOLOGY

This study quantitatively examines the dyadic communication between Subject A and Subject B and their usage of terms of endearment through three dimensions: lexical, temporal, and linguistic, using a predetermined dictionary of terms (see Appendix B).

The paper will first discuss the subjects' usage of terms of endearment from a lexical perspective, focusing on the exact terms that were exchanged in the communication system. This section of the research addresses the following questions: (1) How many messages did the subjects exchange? (2) Which terms of endearment were most frequently used? and (3) At what rate did the subjects utilize terms of endearment?

The second section of the study investigates how the subjects changed their habits of using terms of endearment over the 11 months. This section will study this change by tracking the subjects' number of messages per month and number of times they used terms of endearment per month. This portion will also use a heat graph to represent the subjects' communications' monthly patterns.

Lastly, the paper examines the subjects' use of terms of endearment in English, Chinese (Mandarin), Korean, and French and Spanish. The study looks at (1) the percentage a language takes up in a subjects' speech (2) how these percentages change over time and (3) the subject's linguistic entropy, defined here as the diversity across different language usage.

This paper's data was obtained from the subject's chat history on Instagram, a social media platform launched by Meta. The study uses Python to conduct analysis on the JSON file acquired from Instagram, and finally uses the Matplotlib library for visualization.

4. RESULTS

4.1 Lexical Dimension

The results of the analysis show that the subjects exchanged a total of 8983 messages over the 12 months, with Subject A and Subject B contributing 5041 and 3942 messages respectively (see Appendix A). Within these messages, a total of 138 contained at least one term of endearment, with Subject A sending 78 of them and Subject B sending 60.

Overall, messages with terms of endearment account for 1.54% of the total messages exchanged. On average, Subject A introduces a term of endearment once every 64.63 messages, while Subject B

¹ Also known as "pet names" in informal settings

² The act of switching between languages within a single sentence or clause

introduces one every 65.70. These results suggest a slightly higher frequency of term-of-endearment usage by Subject A.

Data analysis shows that, among both subjects, the 10 most commonly used terms of endearment were *Honey*, *Love*, 오빠, *Baby*, 宝贝, 宝宝, *Sweetie*, 姐姐, *Darling*, and *Sweetheart*, in descending order (see Figure 1).

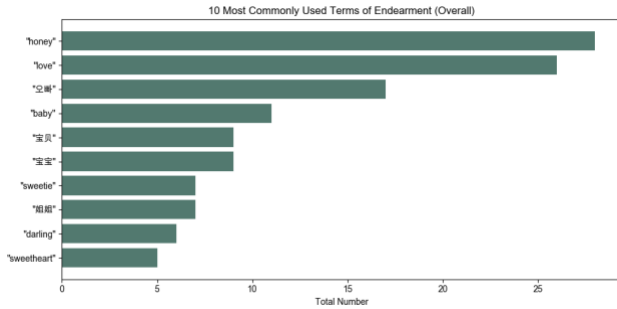


Figure 1: 10 Most Commonly Used Terms of Endearment (Overall)

Figure 2 further categorizes the most frequently used terms based by subject, showing that Subject A's five most commonly used terms were *Honey*, *Love*, 宝宝, 宝贝, and *Sweetie*, while Subject B's five most commonly used terms were 오빠, *Love*, *Honey*, *Baby*, and 아기.

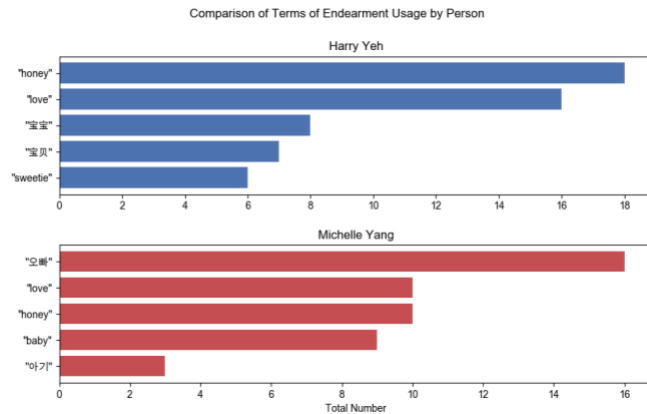


Figure 2: Comparison of Top 5 Most Commonly Used Terms by Subject

The full counts of all terms are listed in Appendix C.

4.2 Temporal Dimension

Figure 3 shows the number of messages exchanged each month between the subjects since June 2025. The graph shows a spike in the number of messages in July 2025, two months into the subjects' relationship, followed by a gradual decrease over the 12 months.

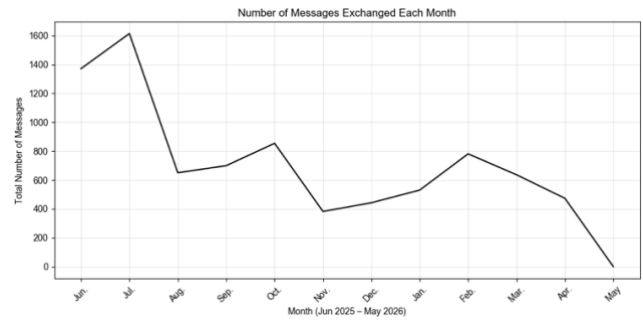


Figure 3: Number of Messages Exchanged Each Month

A similar trend is seen with the number of terms of endearment exchanged by the subjects each month, where both subjects experience a sharp spike in July 2025 (Figure 4). Moreover, the graph shows that while Subject A consistently utilized more terms of endearment from June to October in 2025, Subject B's number of usages surpassed that of Subject A in November 2025, February 2026, and April 2026. Both subjects had a low number of usages in December 2025.

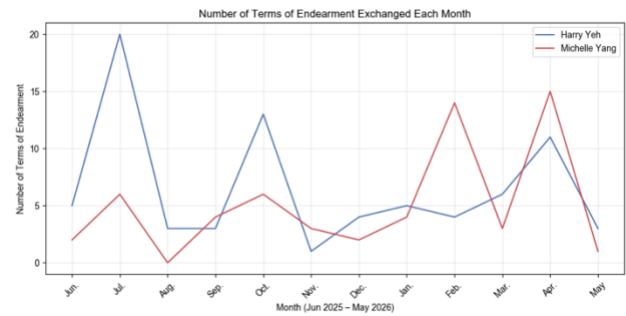


Figure 4: Number of Terms of Endearment Each Month (by Subject)

Lastly, Figure 5 shows a heatmap describing the number of terms of endearment exchanged between the subjects based on the sent month and hour of day. The hours are in Coordinated Universal Time (UTC). The heatmap shows an exceptionally large number of terms of endearment utilized in July 2025 from 00:00 to 01:00, 9:00 to 10:00, and April 2026 23:00 to 24:00. A more consistent pattern of terms being exchanged primarily between 05:00 and 14:00 is seen starting from September 2025.



Figure 5: Messages with Terms of Endearment Based on Sent Month and Hour

4.3 Linguistic Dimension

The last dimension of the study examines how the subjects chose between different languages. This section finds that over time, English terms of endearment have seen a decline in overall proportion (Figure 6): While English was the main or only language used for terms of endearment from June to August 2025, the subjects have begun to use more terms in Chinese and Korean starting from September 2025. In April 2026, English terms only made up less than 40% of all usages of terms of endearment. French and Spanish terms were not used.

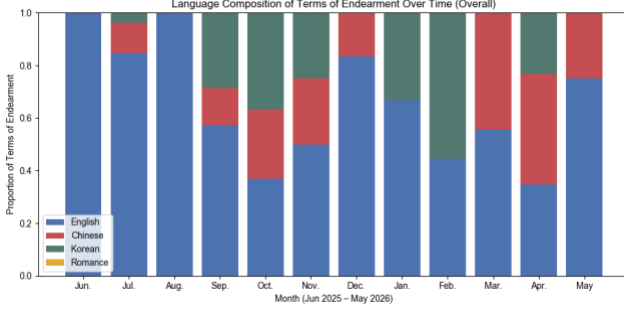


Figure 6: Language Composition of Terms of Endearment Over Time (Overall)

However, as shown by Figure 7, English still remains as the primary language for terms of endearment overall, contributing to 58.70% of all usages, followed by Korean and Chinese. French and Spanish terms are not observed.

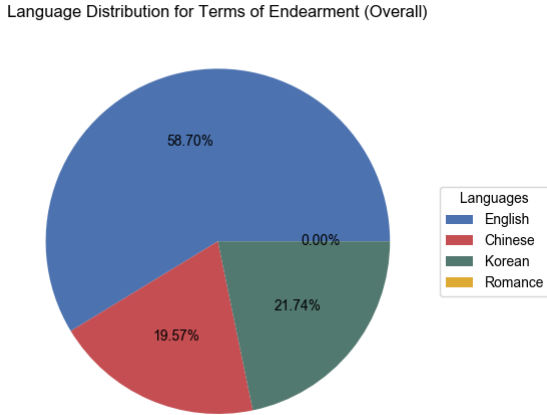


Figure 7: Language Distribution for Terms of Endearment (Overall)

Finally, this portion analyzes the entropy of the two subjects with respect to their language choices. To calculate entropy, the study uses the standard Shannon entropy formulation:

$$H = - \sum_{i=1}^n p_i \log(p_i)$$

Where H is the entropy of the subject's language distribution, p_i is the probability of a term of endearment belonging to category i , and n is the total number of languages examined. In this study, $n=4$ for

the four languages being considered. The logarithm is taken to the natural base unless otherwise specified.

Using this formula, Subject A's entropy of language usage distribution is 0.8789 nats, while Subject B's entropy is 0.9334 nats. The findings suggest that Subject B's usages of terms of endearment are more evenly distributed across the four languages than Subject A. The values of p_i are listed in Appendix D.

5. DISCUSSION

The study's findings show the subject's reactions in the face of schoolwork and long-distance relationships.

In Figure 3, the gradual decline in overall message frequency could suggest a shift in the subjects' communication system, moving from text-based messages to synchronous or in-person interactions. Alternatively, the decline could also be associated with a change in the subjects' communication content, wherein exchanges increasingly consist of audiovisual materials or shared posts rather than written text.

Figure 4 shows a low level of activity for both subjects in December 2025. This period coincides with a time when both subjects were in the same location, making increased in-person conversations a plausible explanation for the comparatively small number of messages in that month. Additionally, both subjects exhibit periodic reduced activities during periods that align with their academic calendars: Subject A demonstrated a substantially low number of messages from November 2025 to January 2026, which lines up with his final examination dates. Similarly, a notable decrease in activity was seen for Subject B in March when her final examinations took place.

The results in Figure 5 show the most activity between the two subjects starting at 5:00 UTC. In Los Angeles, where Subject B is based, this corresponds to 21:00 Pacific Standard Time (PST)³ in the winter or 22:00 Pacific Daylight Time (PDT)⁴ in the spring; in Palaiseau, where Subject A is based, this corresponds to 6:00 Central European Time (CET)⁵ in the winter or 7:00 Central European Summer Time (CEST)⁶ in the spring. Therefore, the increased activity at 5:00 UTC could be explained by the subject's shared activity time given their distinct locations.

Lastly, Figure 6 displays a notable decline in English's proportion in terms of endearment usage from September 2025, along with an increase in Chinese and Korean term usage. A feasible explanation for this pattern relates to Subject B's change in habitat, where in September, she relocated to an environment where multiple people shared a common space. The reduction in English terms of endearment may reflect Subject B's adaptations to moving to a shared living space, where language choice could be impacted by others' presence. This explanation is supported by the revitalization of the English proportion in December and January, when both subjects were in the same location. A second drop happened in February when Subject B was back in the shared space. Hence, the fluctuations in English's proportions in exchanged terms of endearment may reflect a result of Subject B's adaptation to a shared living environment.

6. LIMITATIONS

The Python algorithms that make up the study follow rule-based keyword matching, which does not take context into consideration

³ UTC - 8

⁴ UTC - 7

⁵ UTC + 1

⁶ UTC + 2

when counting and analyzing the data. Therefore, the study can not conclusively exclude cases in which subjects may have mentioned terms of endearment in situations including but not limited to when the word was serving as a non-noun, expressing irony, quoting a third party, or discussing a different subject.

As WeChat – the primary platform for communication between Subject A and B – does not support an official method to export chat history legally, this research was based on the subjects' chat history on Instagram, their secondary platform. The data's source may lead to the samples being smaller or less representative of the subject's actual communication. Additoinally, to ensure the completion of the paper before the actual anniversary, chat history samples only range from June 1st, 2025 to May 20, 2026.

Lastly, the dictionary of terms of endearment was defined by Subject B's subjective interpretations and empirical knowledge over the timespan, which potentially introduces the risk of missing other terms of endearment that were used in the communication sample but missed in the study's analyses.

7. CONCLUSION AND FURTHER WORK

In conclusion, this study conducted numerical analysis on the patterns of Subject A and Subject B's usage of terms of endearment from June 2025 to May 2026. The periodic fluctuations in the quantity of terms of endearment suggest how the subjects adjust the communication system within their significant partnership in the face of academic hardship and distance. The results also note changes in language proportions when the subjects select terms of endearment in different languages, reflecting how the subjects' usage of languages may also be influenced by their changing environments.

The paper lays out a reference framework to analyze dyadic relationships through the perspectives of textual messages and keywords. The methodology of this study proposes a more objective way to use quantitative data to analyze intimate dyadic communication patterns. The study could resolve its limitations to yield more precise results. By introducing Natural Language Processing (NLP) models, the analysis could take conversation context into consideration, excluding mentionings of terms of endearment when the subjects are not referencing each other. Moreover, the results may be more representative if the analyses were based on the subjects' primary platform of communication, allowing the study to work with a larger sample size and more realistic messages and communication styles.

Ultimately, the finds reflect how the everyday language use patterns between Subject A and Subject B serve as subtle evidence of their continued commitment to one another.

8. ACKNOWLEDGMENTS

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Lastly, the author, Subject B, thanks Mr. Sangchun (Harry) Yeh, Subject A, for the fun, care, and happiness he brought to this relationship over the past year. Happy first anniversary, 오빠.

9. REFERENCES

- [1] Sangchun (Harry), Yeh and Michelle Yang. Jun 2025-May 2026. Personal Communication Archive, Instagram Database.

10. CODE AVAILABILITY

The Python code and visualization tools used in this study can be found at: <https://github.com/Michyah/dyadic-affection-analysis>.

APPENDIX A

Dictionary of Terms of Endearment

English Terms of Endearment

- Darling
- Honey
- Sweetie
- Sweetheart
- Dear
- Love
- Baby

Chinese Terms of Endearment

- 宝宝
- 宝贝
- 哥哥
- 姐姐
- 亲爱的

Korean Terms of Endearment

- 오빠
- 누나
- 아기
- 자기야
- Jagiya
- Nuna
- Oppa

French and Spanish Terms of Endearment

- Ma chérie
- Petite amie
- Petite ami
- Mi cariño
- Mi cariña
- Media naranja

APPENDIX B

Number of Messages Every Month

Month (Jun 2025 - May 2026)	Total Messages	Messages with Terms of Endearment
June	1371	7
July	1614	26
August	650	3
September	699	8
October	854	20
November	382	4
December	442	6
January	530	9
February	781	18
March	636	9
April	787	26
May	237	4
Total:	8432	131

APPENDIX C

Terms of Endearment Count by Subject

SUBJECT A		SUBJECT B	
Term	Counts	Term	Counts
Honey	18	오빠	16

Love	16	Honey	10
宝宝	8	Love	10
宝贝	7	Baby	9
Sweetie	6	아기	3
姐姐	6	Darling	2
Darling	4	宝贝	2
Sweetheart	4	哥哥	2
자기야	4	Oppa	2
Dear	2	Sweetheart	1
Jagiya	2	宝宝	1
Baby	2	姐姐	1
아기	1	자기야	1
오빠	1	Sweetie	1
Cutie	0	Cutie	1
哥哥	0	Dear	0
亲爱的	0	亲爱的	0

누나	0	누나	0
Nuna	0	Jagiya	0
Oppa	0	Nuna	0
Ma cherie	0	Ma cherie	0
Petite amie	0	Petite amie	0
Petite ami	0	Petite ami	0
Mi cariño	0	Mi cariño	0
Mi cariña	0	Mi cariña	0
Media naranja	0	Media naranja	0

APPENDIX D

Probability of Subjects' Tendency to Use English, Chinese, Korean, and French or Spanish (p_i)

Subject: Language:	Subject A	Subject B
English	62.8%	53.3%
Chinese	26.9%	10.0%
Korean	10.3%	36.7%
French or Spanish	0%	0%